AI vs. Human: Data-Driven Prediction Model for Brain Stroke Final Report

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Overview

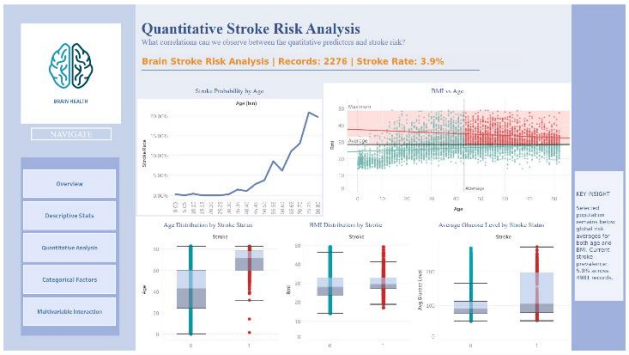
This document captures a comparison between human-driven and AI-driven statistical and machine learning models used for brain stroke data analysis. The human-driven analysis was conducted following a traditional work frame with the help of R and Python. The AI-driven analysis used two different ChatGPT Pro models, the standard GPT model and Data Analyst GPT model specifically developed for data analytics. The models produced and compared include logistic regression, decision tree, and random forest models. Each analytical approach is compared based on different metrics regarding the workflow, running time, visualizations, and statistical results. This comparison investigates the similarities and differences between the two workflows and examines the capabilities of AI to conduct or assist statistical analysis, providing further insight to rapidly increasing AI usage.

1. Data Visualization

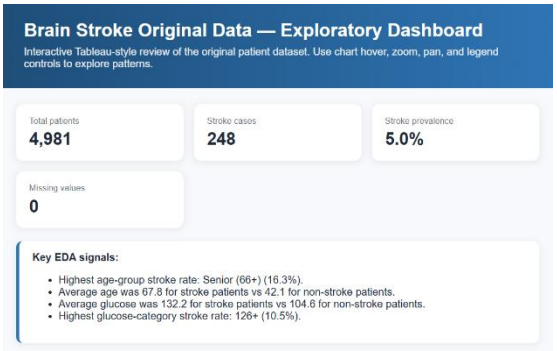
In the human-driven analysis, I used Tableau to build data visualizations and interactive dashboards to perform exploratory data analysis. For the AI-driven analysis, I prompted both standard and Data Analyst models to produce exploratory data visualizations and compose an interactive dashboard. The AI models were able to generate simple graphs and plots capturing linear relationships between two variables, distributions of categorical data, and stroke prevalence heatmaps. Both AI models generated similar graphs using Python programming, however, only the Data Analyst model was able to compose an interactive dashboard with adjustable filters.

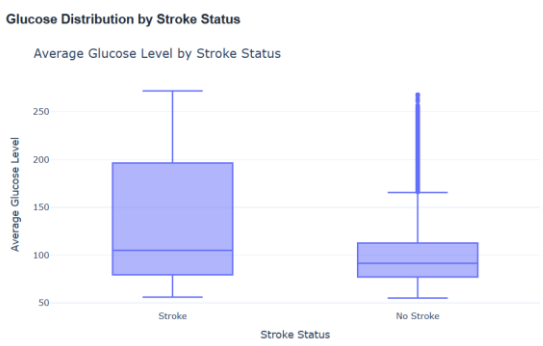
Although the AI models were quicker in making visualizations, they could not complement the complexity and delicacy of human-built Tableau visualizations in terms of multi-variate relationships, variable distributions, statistical insight, and interactivity. This outlines that generative AI may be useful to quickly visualize simple graphs to obtain insight on variable relationships or preview plots in planning stages.

Link to Tableau dashboard: Stroke Data Visualizations | Tableau Public



Link to AI-generated dashboard (Data Analyst model): Brain Stroke Original Data EDA Dashboard





2. Analytical Workflow

In human-driven analysis, the models were built and trained using two different programming languages, R and Python, where the code was manually written based on the analytical requirements. The code is divided into sections, and each code segment was run at each step of the analysis to obtain corresponding results. The interpretations of the results were produced based on human statistical insights and manually kept track on a word document throughout the analysis process.

This workflow required more time, especially when revising code, re-running models, correcting errors, and updating documentation. However, it also provided greater visibility and control over the analytical process. Because the code was structured, tangible, and directly accessible, each step could be reviewed, modified, and reproduced more easily. The researcher could identify where a result came from, trace how a model was built, and verify whether the correct dataset, variables, parameters, and evaluation methods were used. This made the human-driven workflow more reproducible, since both the code and the results were manually tracked and documented throughout the analysis.

The AI-driven analysis also followed the same requirements for datasets and model parameters. However, the analysis and results were entirely automated by AI based on the prompt entered. To improve organization and efficiency, the AI agents were divided into three roles: Prompter, Analyst, and Recorder. The Prompter AI was used to generate structured prompts comprising step-by-step instructions and requirements for the analysis. These prompts were entered into the Analyst AI, which processed the natural language, generated Python code, ran the necessary analyses, and produced results. Finally, Recorder AI was used to organize the prompts and corresponding results into an analysis document.

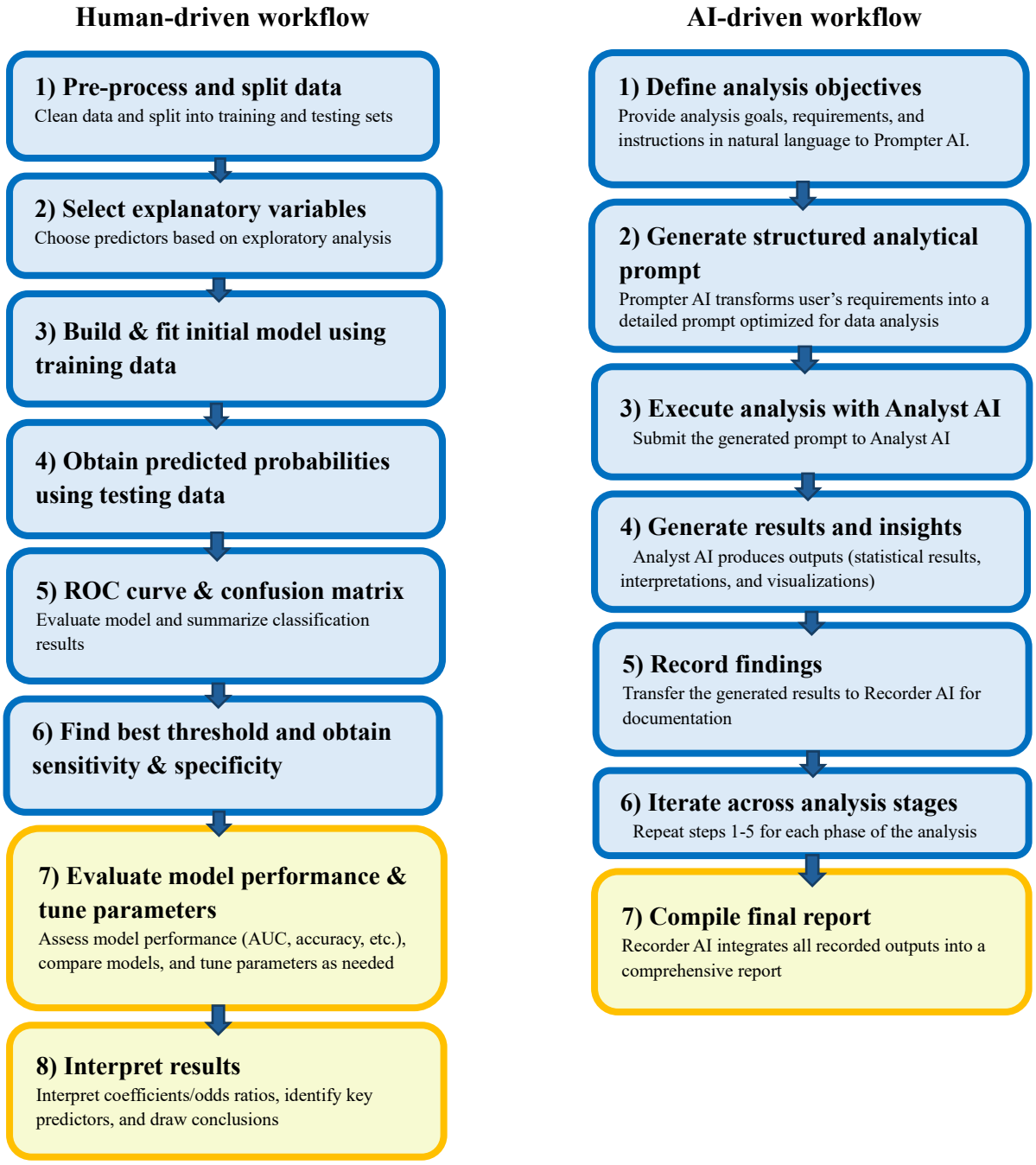
For the Analyst AI, two different ChatGPT Pro models were used: the standard model and the Data Analyst model. The same prompt generated by the Prompter AI was entered into both models, and the results from each were summarized by separate Recorder AI chats. This workflow allowed comparison between the analytical results as well as ensuring the consistency and quality of the outputs from different AI models.

The main advantage of the AI-driven workflow was speed and convenience. It reduced the time required to generate code, run analyses, organize results, and create written summaries. However, the workflow also had less visibility than the human-driven approach

because many steps occurred automatically within the AI system. Although the AI could generate and run code quickly, the analytical decisions, intermediate steps, and possible assumptions were not always as transparent unless they were explicitly requested in the prompt or recorded afterward. Model parameters and methods were also automatically set by the AI unless specified by prompts, which game less control over model settings. This made it harder to verify exactly how some results were produced and increased the need for manual review.

Overall, the human-driven workflow was slower but more transparent and reproducible because the code, outputs, and interpretations were manually controlled and documented. In contrast, the AI-driven workflow was faster and more convenient, but it required stronger oversight to confirm that the correct methods were applied and that the results were interpreted accurately. This comparison suggests that AI is most useful as an assistive tool for code generation, documentation, and workflow acceleration, while human oversight remains necessary for reproducibility, statistical reasoning, and methodological validation.

2.1 Analysis Workflow Flowchart



3. Analyzing & Processing Time

The two approaches are compared based on their analyzing and processing time. In this comparison, these time estimates refer to total workflow time, including setup, prompt/model development, output generation, interpretation, and documentation, rather than only the computer runtime required to train the models.

The overall time required to complete each analytical stage was substantially reduced in the AI-driven workflow. The Prompter AI averaged approximately 1.5 to 2 minutes to generate a complete task-specific prompt. Each prompt was designed to focus on one analytical task. The Analyst AI, using the standard GPT model and Data Analyst model, averaged around 5 to 6 minutes to execute each prompt and generate structured results. The Recorder AI averaged between 2 to 3 minutes in organizing the results and updating the report document. Overall, the total time taken at each step of the AI-driven analysis averaged approximately 8 to 10 minutes. Some timing outliers occurred due to slow internet connection, software delays, or system errors that caused temporary delays and interruptions in the AI performance.

Human-driven Analysis

	Average processing time
Model construction	1 hour
Model training	<1 min (using R and Python)
Interpretation/ Documentation	2 hours
Total analysis time	~3 hours

AI-driven Analysis

	Average processing time (minutes)	**Model(s)**
Prompter	1.7	Standard
Analyst	5.5	Standard/Data Analyst
Recorder	2.5	Standard/Data Analyst
Total analysis time (per prompt)	9.7	

Overall, the use of generative AI substantially shortened the time required to complete comparable stages of the analysis. While the human-driven workflow took an average of 3 to 4 hours for each of the model construction, interpretation, and documentation, the AI-driven workflow completed similar analytical tasks in approximately 1 to 2 hours. Considering solely the total analysis time, the AI-driven analysis outpaced the human-driven analysis by taking approximately 50% less time. This suggests that AI can improve efficiency by accelerating code generation, model setup, result formatting, and report documentation.

However, the reduced time should be interpreted cautiously. The primary increase in efficiency is gained from automation of analysis work frames rather than reducing the computational training time, which was already less than one minute in the human-driven analysis. In addition, the AI-driven workflow time estimates do not fully account for additional time spent verifying outputs, identifying errors, and correcting AI-generated results. These verification time took on average 20 to 30 minutes, sometimes up to an hour, for each result that the AI generated. While the AI-driven workflow was faster overall, its efficiency depended on continued human oversight and validation. The AI-driven workflow was faster in producing outputs, but the human-driven workflow provided greater control over each analytical decision. Therefore, the most meaningful comparison is not speed alone, but speed adjusted for accuracy, reproducibility, and the amount of human validation required.

4. Statistical Results

4.1 Logistic Regression Models

The main-effects model and tuned balanced model were each selected from the human-driven and AI-driven logistic regression analysis respectively. Both models take the full set of predictors (except work type) as explanatory variables and stroke occurrence as dependent variable. The models were constructed using the same training dataset and evaluated with the same testing dataset. The human-driven main-effects model was built as a traditional logistic regression model in R or Python using statistical modeling libraries, designed for statistical inference. The AI-driven tuned balanced model was built in Python using machine learning libraries, mainly scikit-learn, and focused more on prediction performance than statistical inference using tuning mechanisms.

4.1.1 Significant Predictor Comparison

The predictors ranked by their positive contribution to stroke prevalence based on p-values/z-values and odds ratio for both models are shown below. For the AI-driven tuned balanced model, the provided p-values or z-values have been approximated. Because the AI-driven model was tuned and class-balanced for prediction, its p-values should be interpreted cautiously.

Human-driven: Main-effects model

Rank	Predictor	p-value	z-value	Odds ratio
1	Age	<0.001	9.139	1.0764
2	Hypertension	0.006	2.751	1.8479
3	Average Glucose Level	0.016	2.416	1.0041
4	BMI	0.076	1.772	1.0310
5	Smoking Status (Smokes)	0.387	0.865	1.3115

AI-driven: Tuned balanced model

Rank	Predictor	p-value	z-value	Odds ratio
1	Age	<0.001	21.0734	3.1104
2	Average Glucose Level	<0.001	5.1835	1.2239

3	Hypertension	<0.001	3.231	1.1555
4	Heart Disease	0.0010	3.3005	1.1218
5	Smoking Status (Smokes)	0.3784	0.8808	1.0721

The top three predictors of stroke occurrence are consistent across the main-effects model (human-driven) and tuned balanced model (AI-driven). Both models identified age as the most influential predictor by far with the highest statistical significance (smallest p-value) and magnitude of effect (greatest odds ratio). Hypertension remained statistically significant in both models. In the human-driven model, it had a larger odds ratio, suggesting a stronger estimated association with stroke odds within that model. In the AI-driven model, hypertension had a smaller odds ratio but stronger statistical evidence, as shown by its lower p-value. However, because the two models may use different scaling and weighting procedures, the odds ratios should not be directly compared without caution. Average glucose level also remains in the top rank for both models but it appeared more prominent in the AI-driven tuned balanced model. Smoking status appeared in the ranked predictor lists for both models, but it was not a stable statistically significant predictor. This suggests that smoking status is less consistent predictive value in this dataset compared with age, hypertension, and glucose level.

The largest differences between the models involved BMI and heart disease. While BMI showed moderate association in the human-driven model, it had no evidence of significance in the AI-driven tuned balanced model (p-value = 0.5784). In contrast, heart disease was not significant in the human-driven model (p = 0.241) but became a strong predictor in the AI-driven model (p-value = 0.0010). This difference may reflect the effect of class weighting, preprocessing, variable scaling, or the different goals of the two models. The human-driven model is more focused on statistical inference, while the AI-driven tuned balanced model is more focused on improving prediction for the minority stroke class.

Overall, the two modeling approaches show convergence on the most important predictors, especially age, hypertension, and average glucose level. This strengthens confidence that these variables are meaningfully associated with stroke occurrence in the dataset. However, the models diverge for secondary predictors such as BMI, heart disease, and smoking status. These differences suggest that predictor importance depends partly on

modeling choices, including class balancing, preprocessing, and whether the model is designed primarily for inference or prediction. The AI-driven results are useful for identifying predictive patterns, but the human-driven main-effects model may provide a more conventional framework for statistical inference.

4.1.2 Test-set Evaluation Comparison

The accuracy, sensitivity, and specificity using the testing dataset and the best threshold value, are compared below:

	Human-driven: Main-effects Model	**AI-driven: Tuned Balanced Model**
Accuracy	0.7064	0.6929
Sensitivity	0.8560	0.8468
Specificity	0.6985	0.6848
ROC-AUC	0.8270	0.8438
Best Threshold	0.0425	0.4478

Overall, because both human-driven and AI-driven analysis utilized Python algorithmic programming to build and test the models, the resulting models showed similar performances in terms of stroke prediction measured upon accuracy, sensitivity, and specificity. The human-driven model shows slightly higher values in accuracy, sensitivity, and specificity, which may suggest improved suitability for screening. A noticeable difference comes from the best threshold values, which is discussed in section IV (Errors in AI-driven Analysis).

4.2 Decision Tree Models

4.2.1 Model Structure Comparison

The decision tree models selected for comparison are tuned model and class-weighted model from the human-driven and AI-driven decision tree analysis respectively. The models were selected based on their performance. Both models take the full set of predictors as explanatory variables and stroke occurrence as the target variable. The models were constructed using the same training dataset and evaluated using the same testing dataset.

Both models are tuned/pruned versions of an initial default model to improve performance metrics. The human-driven tuned model was built using both R and Python machine learning libraries while the AI-driven tuned balanced model was built using only using Python. The AI-driven used class weight = “balanced” to account for the low stroke rate and increase the model’s attention to stroke cases. The tree model summary is displayed in the table below.

	Human-driven: Tuned model	**AI-driven: Class-weighted model**
Number of nodes	39	7
Number of leaves	20	4
Maximum tree depth	5	2
Minimum samples split	10	2
Minimum samples leaf	5	1
Root node	Agge <= 73.5	Age <= 53.5
CCP Alpha	0.0001	0.0
Training time (seconds)	0.0148	0.0239

The human-driven tuned decision tree and AI-driven class-weighted decision tree show different modeling priorities. The human-driven model is more complex, having 39 nodes, 20 leaves, and a maximum depth of 5, and incorporates multiple predictors at each split. In contrast, the AI-driven model is much shallower, having only 7 nodes, 4 leaves, and a maximum depth of 2, having a much simpler and shallower structure. The human-driven tuned tree provides a more detailed structure and captures more predictor interactions, while the AI-driven class-weighted tree is simpler and easier to interpret but provides less information about predictor relationships.

Age appears at the root node of both decision tree models, with a threshold of <=73.5 for the human-driven model and <=53.5 for the AI-driven model. The split threshold

4.2.2 Variable Importance Comparison

The top five predictors ranked by their variable importance scores of both models are shown below.

Human-driven: Tuned Decision Tree

Rank	Variable	Importance
1	Age	0.4521
2	Average Glucose Level	0.1770
3	BMI	0.1762
4	Hypertension	0.0588
5	Work Type (private)	0.0392

AI-driven: Class-Weighted Decision Tree

Rank	Variable	Importance
1	Age	1.0000
2	Gender (Male)	0.0000
3	Average Glucose Level	0.0000
4	Heart disease	0.0000
5	Ever married (Yes)	0.0000

Both models identify age as the most important predictor with the highest importance scores. However, the AI-driven model appears to use age as the only splitting variable, making it the most important model-associated predictor (importance score = 1) and giving all other predictors zero importance. This suggests that the AI-driven tree functions primarily as an age-based screening model. While this simplicity may reduce overfitting, it may also underfit the data by relying only on age. This supports the importance of age but limits the model's ability to explain how other clinical or demographic factors contribute to stroke prediction.

The human-driven model provides more insight into predictors other than age that may be clinically relevant. Being consistent with the logistic regression model, average glucose level and BMI were identified as leading predictors with moderate variable importance of approximately 17%. The variables next in line, including hypertension and work type had dropped significantly in their scores, suggesting weak relation with stroke prevalence.

4.2.3 Test-set Evaluation Comparison

The accuracy, sensitivity, and specificity using the testing dataset and the best threshold value, are compared below:

	Human-driven: Tuned Tree	AI-driven: Class-weighted Tree
Accuracy	0.6916	0.6736
Sensitivity	0.8240	0.8306
Specificity	0.6846	0.6654
ROC-AUC	0.7880	0.8029
Best Threshold	0.0512	0.6094

In test-set evaluation, the two models show a tradeoff in sensitivity and specificity. The AI-driven decision tree has slightly higher sensitivity, handling class imbalance, but the human-driven model achieved higher accuracy and specificity. This means that while the AI-driven model identified a higher percentage of stroke cases, it produced more false positives. The human-driven model performed better at identifying non-stroke cases. Overall, the AI-driven model may be preferable when sensitivity is prioritized, while the human-driven model provides more balanced and informative results across multiple predictors.

Finally, the best threshold values show a large difference. The AI-driven model's best threshold of 0.6094 is significantly higher than the human-driven threshold of 0.0512, which was the same incidence observed in the logistic regression comparison as well. This may indicate class weighting effects or possible errors that need careful interpretation, which will be addressed in section IV.

4.3 Random Forest Models

4.3.1 Model Structure Comparison

The models selected for comparison show two different modeling strategies. The human-driven model was tuned using balanced class weights, 100 trees, and a maximum tree depth of 6, designed to address class imbalance and limit overfitting. The AI-driven model used 500 trees with no specified maximum depth and no class weighting or other tuning, making it computationally larger and complex. This accounts for the training time of the AI-driven model being substantially longer, but both models were still quick in absolute terms,

so training time is not a major limitation.

Both models take the full set of predictors as explanatory variables and stroke occurrence as the target variable. The models were constructed using the same training dataset and evaluated using the same testing dataset. The human-driven tuned model was built using both R and Python machine learning libraries while the AI-driven tuned balanced model was built using only using Python. The random forest model summaries are displayed in the table below.

	Human-driven: Tuned model	**AI-driven: Default model**
Number of trees	100	500
Maximum tree depth	6	None
Class weighting	Balanced	None
Random state	42	42
Split criterion	Gini Impurity	Gini
OOB Accuracy	0.9592	0.9482
OOB Error	0.0408	0.0518
Training time (seconds)	0.0148	2.3817

Both models used Gini Impurity for splitting criterion and the same random state for a more direct comparison of algorithmic performance. While both models achieved similar out-of-bag accuracy and error, the human-driven model achieved slightly higher accuracy and lower error. However, because the stroke rate in the dataset is only about 5%, OOB accuracy should be considered alongside sensitivity, specificity, and confusion matrix results to

4.3.2 Variable Importance Comparison

The top five predictors ranked by their variable importance scores of both models are shown below.

Human-driven tuned random forest model

Rank	**Variable**	**Importance**
1	Age	0.4892
2	Average Glucose Level	0.1272

3	BMI	0.1054
4	Hypertension	0.0613
5	Ever married (Yes)	0.0562

AI-driven random forest model

Rank	Variable	Importance
1	Average Glucose Level	0.2446
2	Age	0.2437
3	BMI	0.2232
4	Hypertension	0.0302
5	Heart disease	0.0269

The variable importance results show both agreement and difference between the models. Both models identified the same set of predictors, age, average glucose level, and BMI, to be in the top three rank, suggesting that these variables were consistently used for prediction. However, the human-driven model identified age as the strongest model-associated predictor with importance = 0.4892, which is also consistent with the results of the previous models. The following predictors, including average glucose level, bmi, and hypertension, have significantly smaller importance compared to age. On the other hand, the AI-driven model identified average glucose level as the strongest predictor, but the importance is more evenly distributed across age and BMI, showing similar importance level. The models agree that age, average glucose level, and BMI contributed the most to prediction performance, but this does not directly imply that they have clinical causation, statistical significance in association with stroke risk.

4.3.3 Test-set Evaluation Comparison

The accuracy, sensitivity, and specificity using the testing dataset and the best threshold value, are compared below:

	Human-driven: Tuned model	**AI-driven: Default model**

Accuracy	0.7020	0.7110
Sensitivity	0.8080	0.8145
Specificity	0.6964	0.7055
ROC-AUC	0.8120	0.8006
Best Threshold	0.0721	0.0420

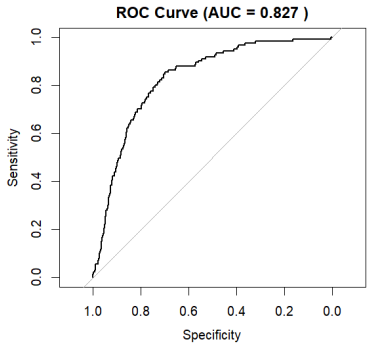
Both models showed similar achievements in terms of accuracy, sensitivity, and specificity, with the AI-driven model achieving slightly higher metrics. This implies that the AI-driven model misses fewer stroke cases while also creating fewer false positive alarms, suggesting better threshold specific classification. However, the human-driven model had a slightly higher ROC-AUC, indicating that the overall ability to separate stroke and non-stroke cases across possible thresholds is higher. The AI-driven model performed slightly better at the selected threshold, while the human-driven model showed better overall ranking ability and used a more controlled structure designed for the imbalanced dataset.

Both selected thresholds are low, which is expected in a dataset with low stroke prevalence. The best threshold values are apart by approximately 0.03, which is not a huge difference compared to the discrepancies observed in the previous models.

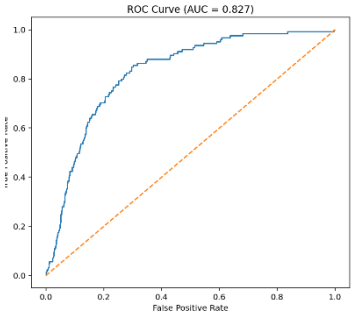
5. Error in AI-driven analysis

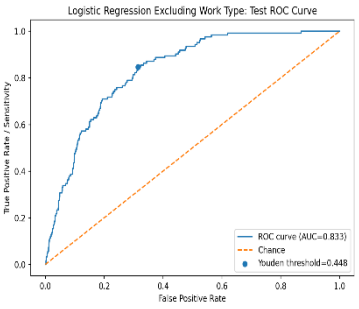
A key discrepancy between the human-driven and AI-driven test-set evaluation results was observed in the selected classification threshold. In the logistic regression analysis, both workflows selected the best threshold using Youden's Index from the ROC curve. However, the resulting thresholds differed substantially: the human-driven model identified a threshold of 0.0425, while the AI-driven model reported 0.4478. Recalculating the AI-reported threshold in both Python and R, sensitivity dropped to 0.0160 and specificity increased to 0.9953. This indicates that the model classified almost all observations as non-stroke, which is inconsistent with the purpose of Youden's index, seeking best balance between sensitivity and specificity. The ROC curve plots of both work frames further outline where the error was made in the AI-driven analysis:

Main-effects Model



Tuned balanced Model





The error appears to come from a misinterpretation of the ROC curve. An ROC plot shows sensitivity, or true positive rate, against false positive rate. The ROC curve for the AI-built tuned balanced model appears to show a point around x = 0.4478, which represents a false positive rate of approximately 0.4478. Therefore, interpreting this value as the model's best classification threshold is incorrect.; Youden's index is calculated as sensitivity + specificity -1, which is equivalent to the difference between true positive rate and false positive rate. Graphically, this means Youden's Index selects the ROC point with the greatest vertical distance above the no-discrimination line. However, the x-coordinate of that selected point is still the false positive rate, not the threshold used to classify observations. The actual threshold must be obtained from the threshold array corresponding to the maximum Youden index value. Therefore, 0.4478 should not be treated as a threshold value, and thus this reflects an error in ROC curve interpretation and threshold extraction in the AI-driven analysis.

A similar issue reappeared in the AI-driven decision tree analysis, where the x-coordinate of the ROC curve was treated as the selected threshold, resulting in an unusually high cutoff. This suggests that the error was not isolated to one model, but reflected a repeated conceptual misunderstanding in the AI-driven workflow.

6. Interpretability & Reproducibility

Overall, the human-driven workflow showed higher interpretability and consistency across model results. The AI-driven workflow was useful for quick generation and execution of code, which reduced the time required for building and evaluating complex statistical and machine learning models. However, the AI-driven analyses also required frequent manual

reviews and corrections. When errors occurred, they were difficult to identify because there was no direct access to the code, which still ran successfully while containing conceptual or statistical mistakes. This raised concerns about relying on AI-generated analysis without independent verification. This also questions the AI's efficiency and reliability in spite of its fast automative abilities.

The AI-driven workflow was also more dependent on prompt specificity. To obtain consistent and accurate result, the prompts needed to define the purpose, model settings, evaluation metrics, threshold-selection procedure, etc. in detail. This shifted the analytical workload from actual model construction/training to designing prompts and validating outputs. Based on this study, AI appears most useful as an assistive tool for code generation, workflow organization, and documentation, rather than as a fully independent analytical system. Human review remains necessary to verify statistical reasoning, confirm reproducibility, and ensure that model results are interpreted correctly. The main limitation of the AI-driven workflow was not speed or code generation, but statistical reliability. The AI could generate working code efficiently, but it still required human oversight to confirm that the methods were applied and interpreted correctly.